

Question Paper

Exam Date & Time: 24-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

FIFTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

SUBJECT: CSE 3122/CSE_3122 SOFTWARE ENGINEERING

(SPL: COMPUTER SCIENCE & ENGINEERING)

Marks: 50

Duration: 180 mins.

Answer all the questions.

- 1A) Apply your knowledge of Systems Engineering to demonstrate how self-driving cars integrate hardware and software components to achieve autonomous functionality with a suitable diagram. Identify the activity that must be carried out throughout the system development process and apply the same concept to propose two other examples of product development projects where Systems Engineering can be effectively implemented. (5)

Course Outcome: "CO1" / Cognitive Level: "Application"

- 1B) Identify which Software Development Life Cycle (SDLC) model to be used in the development of WhatsApp. Explain the life cycle activities of this Model with a diagram. (3)

Course Outcome: "CO1" / Cognitive Level: "Application"

- 1C) Differentiate pair programming in XP from traditional programming with a suitable example. (2)

Course Outcome: "CO1" / Cognitive Level: "Application"

- 2A) In a loan processing application with modules like "Verify Applicant", "Calculate Eligibility", and "Disburse Loan", developers face repeated data validation and unclear module interfaces. As the design lead, explain how you would redesign the system using structured design principles to minimize redundancy and ensure clear communication among modules by highlighting the roles of cohesion and coupling in your solution. (5)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- 2B) The SRS for a new banking application is heavily technical and filled with IT Professionals, making it difficult for non-technical clients to review. Explain how you would modify the SRS structure or language to make it clear and understandable for stakeholders with limited technical knowledge. (3)

Course Outcome: "CO2" / Cognitive Level: "Application"

- 2C) Imagine your organization is refactoring a legacy payroll application, your team must choose between improving cohesion first or reducing coupling. Evaluate which should be prioritized and justify your decision. (2)

Course Outcome: "CO3" / Cognitive Level: "Evaluate"

- 3A) A Smart Travel Recommendation System provides personalized trip suggestions to users. Each user has a profile with travel preferences (budget, location type, duration). The system maintains a

database of destinations, hotels, and activities. Based on user input, it generates travel packages using an AI-based recommendation engine. Travel agents can upload new package details, and the admin maintains user and booking records. Draw a Class Diagram for this system showing key classes (User, Destination, Package, Hotel, Activity, Admin, RecommendationEngine), their attributes, operations, and relationships.

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 3B) A Healthcare Appointment Scheduling System lets patients book appointments with doctors, view schedules, and receive reminders. Receptionists manage patient records and doctor availability. Draw a Hierarchy Chart (Structured Design) for this system, showing main modules and submodules. (3)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 3C) Analyze the use of formal methods in software development by examining how their two primary merits and two major limitations influence the overall software design and quality assurance process. (2)

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- 4A) Analyse the concept of postscripts to explain how they can be used to improve software engineering processes in real-world projects. (5)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 4B) Analyze the workflow of a Hospital Management System by examining how Appointment Management, Patient Record Management, and Billing & Inventory Management interact within the system. Using your analysis, design and draw an Activity Diagram that clearly represents the flow of activities across these three components. (3)
- Appointment Management, Patient Record Management and Billing and Inventory Management.

Course Outcome: "CO4" / Cognitive Level: "Application"

- 4C) Analyze the state transitions involved in a movie theatre ticket-booking process and use your analysis to design and draw a State Chart Diagram for the Booking object, showing the states Created, Pending Payment, Confirmed, and Cancelled, along with the events that trigger transitions between them. (2)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 5A) Analyze the concept of White Box Testing and discuss various coverage-based testing strategies used in it with suitable examples. (5)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

- 5B) Apply various program debugging techniques to identify and resolve software errors that occur during the development process. (3)

Course Outcome: "CO5" / Cognitive Level: "Application"

- 5C) Analyze the concept of Security Testing and illustrate how it helps identify system vulnerabilities with a suitable example. (2)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

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Event Name	Subject
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#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	5
3	3	CO 3	10
4	4	CO 4	15
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025 CSE_3122:Software Engineering

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	8	24(48%)
4	Analysis	6	24(48%)
5	Create	1	2(4%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)